Self Study Exam

Q 1:

$0.23 + 0.21 + 0.25 + 0.25 = 1$

$P(x_i) \geq 0 \rightarrow$ True

$P(x_i) = 1 \rightarrow$ True

True

Q 2:

$0.1 + k + 0.2 + 2k = 1$

$0.3 + 3k = 1$

$\frac{3k}{3} = \frac{0.7}{3}$

$\boxed{k = \frac{7}{30}}$

$0.1 + \frac{7}{30} + 0.2 + 2\left(\frac{7}{30}\right) = 1$

Q 3:

$P(x \geq 2) \rightarrow 2, 3, 4$

$P(x = 2) + P(x = 3) + P(x = 4)$

$\frac{2}{10} + \frac{3}{10} + \frac{4}{10} = \frac{9}{10}$

Q 4 :

$S = \{ BB, GB, GB, GG \}$ $\qquad$ $X = 0, 1, 2$

X	0	1	2
$P(X)$	$1/4$	$2/4$	$1/4$
X^2	0	1	4

Mean :

$$E(X) \text{ or } M = x_1 P(x_1) + x_2 P(x_2) + \dots + x_n P(x_n)$$

$$= 0(1/4) + 1(2/4) + 2(1/4)$$

$$= 1$$

$$E(X^2) \text{ or } M^2 = 0(1/4) + 1(2/4) + 4(1/4)$$

$$= 2/4$$

Variance :

$$var(X) = E(X^2) - [E(X)]^2$$

$$= 3/2 - (1)^2$$

$$= 1/2$$

$$S.D = \sqrt{1/2} = 0.7071$$

Sondos Yaqoob Al Rahbi

16J20145